

Adsayan Prabaharan

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Profile

Mathematics undergraduate at Imperial College London with a strong foundation in analytical problem-solving and computational methods. Achieved 4 A*s at A-level and demonstrated mathematical excellence through three different Olympiad awards. Proficient in Python and R, with experience in algorithms and data-driven thinking developed through Harvard's CS50. Particularly interested in the intersection of mathematics, computing, and quantitative finance.

Education

Imperial College London – MSci Mathematics September 2025 – July 2029

- Societies: Poker Society, Mathematics Society, Algorithmic Trading Society, Cauchy Capital, Investment Society
- Relevant modules: Linear Algebra and Groups (72% average), Introduction to Applied Mathematics (75% average), Probability and Statistics, Calculus and Applications

Queen Elizabeth's School, Barnet September 2019 – June 2025

- Grades: A*A*A* in Mathematics, Further Mathematics and Physics.
- Extra curriculars: Achieved distinction in the British Mathematical Olympiad, Medalist in Maclaurin Olympiad (UKMT), Medalist in Hamilton Olympiad (UKMT). Improved novel problem solving ability as a result of preparing for these competitions.
- Grades: 9 GCSEs at Grade 9, 2 GCSEs at Grade 8

Experience

Mathematics Tutor, Queen Elizabeth's School – Barnet, UK Sep 2023 – Mar 2025

- Tutored students who achieved a gold certificate or above in UKMT challenges. Worked closely with students to support them when solving challenging Olympiad-style problems.

Imperial x LSE Estimathon, Imperial College London, UK Feb 2026

- Participated and placed well in an estimathon, comprising of a variety of Fermi problems. Developed a strong reasoning ability as a result of participation.

Cauchy Capital Axioms, Imperial College London, UK Sep 2023 – Mar 2025

- Attended a series of lectures detailing the algorithmic trading process, as well as how implement various algorithms in Python using the Pandas library.

Industry Insight Day, Howden Re, UK Sep 2026

- Was selected out of 500 applications for an insight programme at Howden Re. Took part and was commended for performance in challenges involving modelling risk profiles of insurance firms. Improved my ability to work quickly on unfamiliar problems under pressure.

Mathematics Ambassador, Imperial College London, UK Sep 2026

- Led a problem-solving session for incoming first year mathematics students at Imperial College London. Developed communication skills and the ability to explain complex problems simply.

Current projects

- **Binomial Option Pricing and Volatility Engine** – Built a Binomial Option Pricing Engine in Python using Cox-Ross-Rubinstein parameterisation to price European and American calls/puts. Developed an understanding of option pricing and improved proficiency of implementation of theoretical models into Python.
- **Independent Research Project** – Researched the long term behaviour of binary quantum systems and presented findings to a panel at Imperial College London. Developed my ability to apply theoretical linear algebra to more real-world contexts.